

Environmental Station Dashboard Startup Procedure

1 Purpose

This document describes the repeatable startup procedure for running the Environmental Station Dashboard pipeline.

The full data path is:

XIAO ESP32S3 → Mosquitto MQTT broker → Node.js dashboard server → browser dashboard

The most important detail is that the laptop IP address may change depending on the network. Therefore, every time the project is started, the current laptop IP address must be checked and used consistently.

2 Important Rule

The same laptop IP address must be used in all MQTT-related places:

- Mosquitto broker address
- Dashboard `server.js`
- Arduino/XIAO firmware
- Optional fake publish test

For example, if the laptop IP is:

```
192.168.0.3
```

then the dashboard and ESP32 must both use:

```
192.168.0.3:1883
```

3 Step 1: Find the Laptop's Current WiFi IP Address

Open PowerShell and run:

```
ipconfig
```

Find the active WiFi adapter and look for the IPv4 address:

```
IPv4 Address . . . . . : 192.168.x.x
```

Example:

```
IPv4 Address . . . . . : 192.168.0.3
```

This IP address is the MQTT broker IP for the current session.

4 Step 2: Update the Dashboard Server IP

Go to the dashboard folder:

```
cd C:\Users\orest\Documents\Arduino\GRAD_PROJECT\dashboard
```

Update the IP address inside `server.js`. For example, if the current laptop IP is 192.168.0.3, run:

```
(Get-Content .\server.js) -replace '192\.168\.d+.d+', '192.168.0.3' | Set-Content .\server.js
```

Then confirm the change:

```
Select-String -Path .\server.js -Pattern "192.168"
```

Expected result:

```
const MQTT_BROKER = 'mqtt://192.168.0.3:1883';
```

If the laptop IP is different, replace 192.168.0.3 with the current IP from `ipconfig`.

5 Step 3: Update the Arduino/XIAO Firmware

Before uploading the firmware to the XIAO ESP32S3, update the MQTT broker IP in the Arduino code.

Find the line:

```
#define MQTT_BROKER_IP "192.168.0.3"
```

Change it to match the current laptop IP address.

Example:

```
#define MQTT_BROKER_IP "192.168.1.27"
```

Then upload the firmware to the XIAO ESP32S3.

6 Step 4: Start the Mosquitto MQTT Broker

Open PowerShell as Administrator and run:

```
cd "C:\Program Files\mosquitto"  
.\mosquitto.exe -c "C:\Program Files\mosquitto\mosquitto.conf" -v
```

Expected output:

```
Opening ipv4 listen socket on port 1883.  
mosquitto version 2.1.2 running
```

Leave this PowerShell window open. This window is running the MQTT broker.

6.1 If Port 1883 Is Already in Use

If Mosquitto prints:

```
Error: Only one usage of each socket address is normally permitted.
```

then another Mosquitto process is already using port 1883.

Run:

```
netstat -ano | findstr :1883
```

Look for the line with LISTENING. Example:

```
TCP 0.0.0.0:1883 0.0.0.0:0 LISTENING 21508
```

The last number is the process ID, also called the PID. Check what process it is:

```
tasklist /FI "PID eq 21508"
```

If it is mosquitto.exe, stop it:

```
taskkill /PID 21508 /F
```

Then start Mosquitto again:

```
.\mosquitto.exe -c "C:\Program Files\mosquitto\mosquitto.conf" -v
```

7 Step 5: Start the Dashboard Server

Open a second PowerShell window and run:

```
cd C:\Users\orest\Documents\Arduino\GRAD_PROJECT\dashboard  
node server.js
```

Expected output:

```
[BOOT] Environmental Station Dashboard server starting...  
[HTTP] Dashboard on http://0.0.0.0:3000  
[MQTT] Connected to mqtt://192.168.0.3:1883  
[MQTT] Subscribed to sensors/xiao01/readings
```

The IP shown in the MQTT line must match the laptop IP found from `ipconfig`. Then open the dashboard in a browser:

```
http://localhost:3000
```

The dashboard should show a green connected status.

8 Step 6: Optional Fake Publish Test

This step tests the laptop-side pipeline without the ESP32.

Open a third PowerShell window and go to the dashboard folder:

```
cd C:\Users\orest\Documents\Arduino\GRAD_PROJECT\dashboard
```

Create a test JSON payload:

```
Set-Content .\payload.json '{"device_id":"xiao_esp32s3_01","sequence":999,"timestamp":"2026-05-27 10:04:00","temperature_c":23.1,"humidity_rh":47.2,"co2_ppm":630,"voc_index":120,"temp_sources":2,"hum_sources":2}' -Encoding ascii
```

Publish the payload using Mosquitto's publisher:

```
Get-Content .\payload.json -Raw | & "C:\Program Files\mosquitto\mosquitto_pub.exe" -h 192.168.0.3 -p 1883 -t sensors/xiao01/readings -s
```

If the current laptop IP is not 192.168.0.3, replace it with the correct IP.

Expected output in the dashboard server window:

```
[DB] Stored: seq=999 T=23.1 RH=47.2 C02=630 VOC=120
```

The browser dashboard should also update with the fake values.

9 Step 7: Run the XIAO ESP32S3

After the dashboard pipeline is confirmed, connect the XIAO ESP32S3 and open the Serial Monitor at:

```
115200 baud
```

The expected successful MQTT publish message is:

```
DEBUG : Record published to sensors/xiao01/readings
```

At the same time, the dashboard server should print:

```
[DB] Stored: seq=...
```

The browser dashboard should update with real sensor data.

10 Common Problems

10.1 Dashboard Server Tries the Wrong IP

Problem:

```
[MQTT] Error: connect ETIMEDOUT 192.168.0.4:1883
```

Cause: `server.js` still contains an old laptop IP.

Fix: run `ipconfig`, get the current laptop IP, and update `server.js` again.

10.2 PowerShell Blocks npx

Problem:

```
npx.ps1 cannot be loaded because running scripts is disabled
```

Fix: use `npx.cmd` instead of `npx`, or avoid `npx` and use `mosquitto_pub.exe`.

10.3 Malformed JSON

Problem:

```
[MQTT] Malformed JSON, skipping
```

Cause: PowerShell destroyed the quotes in the inline JSON.

Fix: use a `payload.json` file and publish through stdin using `mosquitto_pub.exe`:

```
Get-Content .\payload.json -Raw | & "C:\Program Files\mosquitto\mosquitto_pub.exe  
" -h 192.168.0.3 -p 1883 -t sensors/xiao01/readings -s
```

10.4 Mosquitto Port Already in Use

Problem:

```
Only one usage of each socket address is normally permitted.
```

Fix:

```
netstat -ano | findstr :1883  
tasklist /FI "PID eq THE_PID"  
taskkill /PID THE_PID /F
```

Then start Mosquitto again.

11 Short Checklist

Every time the project is started:

1. Run `ipconfig`.

2. Copy the laptop's current WiFi IPv4 address.
3. Put that IP in `server.js`.
4. Put that IP in the Arduino firmware as `MQTT_BROKER_IP`.
5. Start Mosquitto.
6. Start `node server.js`.
7. Open `http://localhost:3000`.
8. Optional: run fake publish test.
9. Run the XIAO ESP32S3.

12 Final Notes

The laptop IP may change between networks. This is normal. The project still works as long as the current laptop IP is used consistently in the dashboard server, the fake publish test, and the ESP32 firmware.